

Sriram Rao Chennamaneni

sriramrao.chennamaneni@gmail.com | USA | 786-464-7511 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building and deploying production applications, backend APIs, cloud services, and data-intensive systems using Python, TypeScript, React, Node.js, SQL, and AWS. Experienced in full-stack development, backend and database systems, cloud architecture, CI/CD, debugging, performance optimization, and production reliability across energy, research, and financial-services environments.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash

Frontend: React, HTML/CSS, Chart.js

Backend & APIs: Node.js, Express, FastAPI, Flask, REST APIs, Swagger/OpenAPI

Cloud & Databases: AWS (Lambda, API Gateway, DynamoDB, S3, CloudFront, EC2, Cognito, IAM, CloudWatch, EventBridge), GCP, MySQL, DB2, SQLite, Data Modeling

DevOps & Systems: Docker, Git/GitHub, GitHub Actions, CI/CD, Terraform, Linux, AWS IoT Greengrass, Modbus TCP

Enterprise & Tools: Control-M, Jira, ServiceNow, COBOL, JCL

PROFESSIONAL EXPERIENCE

Software Engineer – Machine Learning | NextNRG, Inc. | Miami, FL

Jun 2024 – Present

- Built and deployed a full-stack, edge-to-cloud energy management platform across 3 site deployments using React, TypeScript, Node.js, Python, Docker, and AWS for real-time monitoring and operational control.
- Developed 10+ responsive React/TypeScript views for asset monitoring, energy-flow visualization, forecasts, historical analytics, alerts, configuration, and multi-site operations.
- Designed 20+ REST APIs and backend services for real-time telemetry, forecasting, controller state, site configuration, analytics, and dispatch workflows across cloud and edge environments.
- Engineered 8+ serverless AWS workflows with Lambda, API Gateway, DynamoDB, S3, Cognito/IAM, and CloudWatch for authenticated APIs, data processing, forecast delivery, and operational monitoring.
- Built offline-resilient data workflows using scheduled cloud processing, DynamoDB/S3 persistence, and local SQLite caching to maintain critical forecast and operational data during connectivity interruptions.
- Improved production reliability through API/input validation, caching, retryable sync queues, health checks, structured logging, and Dockerized deployments across ARM64 edge and AMD64 cloud environments.

Software Research Assistant | FIU Applied Research Center | Miami, FL

Mar 2023 – May 2024

- Built automated Python pipelines to transform ADMS grid data into OpenDSS-compatible datasets, reducing manual data preparation by 30%.
- Developed and executed automated test cases using TestComplete to validate ADMS functionality and support regression testing for Florida Power & Light workflows.
- Built Python data-processing and ingestion workflows using pandas, NumPy, SQL, Bloomberg, Refinitiv, and WRDS for large-scale energy, financial, and climate datasets.

Junior Software Engineer | Cognizant (Client: Northern Trust) | Hyderabad, India

Jan 2021 – Dec 2022

- Developed and maintained enterprise backend applications and financial data workflows using Python, SQL, DB2, COBOL, and JCL, supporting asset, institution, and pricing data across interconnected Northern Trust systems.
- Built Python and SQL automation scripts for data processing, validation, and reconciliation and optimized database operations and Control-M batch workflows, reducing processing runtime by up to 34%.
- Delivered 20+ application enhancements through coding, unit/integration testing, code review, Jira-based Agile workflows, environment setup, and controlled production releases using ServiceNow.
- Resolved 95% of assigned production issues within SLA through code debugging, SQL/data analysis, log investigation, and root-cause analysis; monitored scheduled workloads and performed post-deployment validation.

SELECTED PROJECTS

Data Engineering Platform | Python, Docker, GCP, Terraform, BigQuery, dbt, Spark, Kafka

- Built batch and streaming data pipelines with Spark and Kafka and deployed cloud infrastructure on GCP using Terraform and BigQuery.

AI Workflows with n8n | OpenAI, Pinecone, Gmail, Google Drive, Tavily

- Built a RAG application using OpenAI embeddings and Pinecone vector retrieval, integrating automated AI workflows for document retrieval and response generation and reducing response time by 50%.

EDUCATION & CERTIFICATIONS

M.S. in Computer Science | Florida International University, Miami, FL

2024

B.Tech. in Computer Science and Engineering | HITAM, India

2020

Certifications: Google Cloud Professional Data Engineer | Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft Certified: Azure Data Fundamentals (DP-900)